

PRACTICAL 5

Aim : Determine whether following IPv4 address are valid or invalid. If valid IPv4 address then find class, Network and Host ID of an IPv4 address. If invalid IPv4 address then write reason for the same.

- a) 298.27.5.6
- b) 1.4.5.5
- c) 11100010.23.14.67
- d) 192.226.12.11.9

Input :

Sr. No.	Address	Valid / Invalid	Class	Network ID	Host ID	Reason
a)	298.27.5.6	Invalid	—	—	—	First octet 298 > 255, so it is not a valid IPv4 address.
b)	1.4.5.5	Valid	Class A	1	4.5.5	All 4 octets are in the range 0–255.
c)	11100010.23.14.67	Invalid	—	—	—	First octet 11100010 is not a valid decimal octet. If it was intended as binary, it would need to be converted to decimal first.
d)	192.226.12.11.9	Invalid	—	—	—	Contains 5 octets; IPv4 addresses must contain exactly 4 octets.

Explanation of the valid address

b) 1.4.5.5

- First octet = 1
- Since the first octet is 1–126, it belongs to Class A.
- Default Class A structure: Network.Host.Host.Host
- Therefore:
 - Network ID = 1
 - Host ID = 4.5.5